

# Rectangular matrix multiplication from polynomial products

Przemysław Uznański\*

Pathway

[przemek@pathway.com](mailto:przemek@pathway.com)

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## Abstract

In this note, we extend the matrix-multiplication analysis in [6] to rectangular products and prove

$$\omega(1, k, 1) \leq \begin{cases} 2, & 0 \leq k \leq \frac{1}{2}, \\ 1 + k + \frac{1}{4k}, & k \geq \frac{1}{2}. \end{cases}$$

In particular,  $\omega(1, \frac{1}{2}, 1) = 2$  and the dual exponent satisfies  $\alpha \geq \frac{1}{2}$ . The proof uses explicit degenerations of convolution tensors, multinomial counting, and Strassen’s spectral theorem. An angular constraint on three dot-product parameters gives both the rectangular curve and a bound for  $\omega(a, b, c)$ , with exact exponent  $a + b$  whenever  $a \geq b \geq c \geq 0$  and  $c(a + b) \leq ab$ . As an application, Zwick’s algorithm [13] for all-pairs shortest paths in directed unweighted graphs runs in  $O(n^{2.5})$  time. Combining the rectangular bound with the  $(\min, +)$ -product improvement of Alman and Vassilevska Williams [1] further gives  $O(n^{2.4999})$  running time.

## 1 Introduction

The motivation for this note is to gather, in one place, useful bounds and algorithmic consequences of the groundbreaking matrix-multiplication result in [6], with a focus on rectangular products.

**Complexity convention.** Throughout this note, all complexity bounds suppress a  $+o(1)$  term in the exponent:  $O(n^c)$  denotes  $n^{c+o(1)}$ . All shape parameters are fixed.

Rectangular matrix multiplication is a basic ingredient in algorithms for all-pairs problems, including shortest paths, lowest common ancestors, and Hamming distance. Its exponents depend on the shape of the product; an improvement for square matrices need not give a comparable improvement for every rectangular shape. The analysis in [6] establishes the square bound  $\omega \leq 2.25$  using a shared-leg entropy inequality and degenerations of convolution tensors. We extend that analysis to rectangular products by retaining the orientation information lost under full leg symmetrization.

We work over  $\mathbb{C}$ . The resulting exponent bounds hold over every field of characteristic zero by invariance of asymptotic tensor rank under field extension [7, Theorem 3.10].

For fixed nonnegative real  $u, v, w$ , let  $\omega(u, v, w)$  be the infimum of the exponents  $e$  for which multiplying an  $\lceil n^u \rceil \times \lceil n^v \rceil$  matrix by a  $\lceil n^v \rceil \times \lceil n^w \rceil$  matrix uses  $O(n^e)$  arithmetic operations, with the convention above. Write  $\omega(k) = \omega(1, k, 1)$  and  $\alpha = \sup\{k \geq 0 : \omega(k) = 2\}$ .

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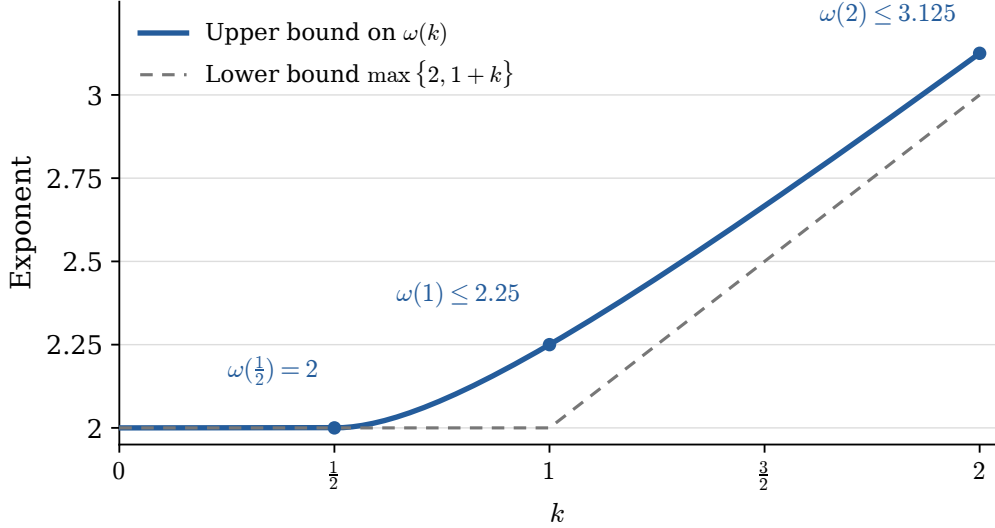


Figure 1: The upper bound on  $\omega(k)$  from Theorem 1.1 for  $0 \leq k \leq 2$ , together with the size lower bound  $\max\{2, 1+k\}$ . The bounds coincide for  $0 \leq k \leq \frac{1}{2}$ .

**Theorem 1.1.** *For every fixed real  $k \geq 0$ ,*

$$\omega(k) \leq \begin{cases} 2, & 0 \leq k \leq \frac{1}{2}, \\ 1 + k + \frac{1}{4k}, & k \geq \frac{1}{2}. \end{cases} \quad (1)$$

Consequently  $\omega(k) = 2$  for  $0 \leq k \leq \frac{1}{2}$ ,  $\alpha \geq \frac{1}{2}$ , and  $\omega(2) \leq 3.125$ .

Theorem 1.1 gives exponent two for inner dimension up to  $n^{0.5}$ , and an upper bound within  $\frac{1}{4k}$  of the input-size lower bound  $1+k$  for  $k \geq 1$ .

The proof has two parts. First, we list explicit ways to split and combine polynomial products and apply a tensor character to them. A character assigns numerical sizes that add on direct sums, multiply on tensor products, and cannot increase under restriction. Multinomial counting gives concavity and growth inequalities for these sizes. Ordinary scaling limits, common slopes, and intercept bounds then give  $\sum_i \arccos \sqrt{p_i} \geq \pi/2$  for the character's three dot-product parameters.

Second, we optimize this constraint and apply Strassen's spectral theorem: the largest character value equals asymptotic tensor rank. We state precisely the form of this theorem we use, then show how the rank bound gives a running-time bound for every fixed matrix shape.

**Relation to the preceding analysis.** The finite separation construction and the two polynomial splittings in Lemma 2.1 come from Proposition 3.1 and the proofs of Lemmas 4.1 and 4.2 of [6]. We recap their constructions in the appendices. Multinomial counting gives the power-sum inequality needed here directly from finite separation. The Hermite direct-sum degeneration, its ordinary scaling limits, and the coupling of the six intercept ratios extend the preceding analysis. Strassen's spectral theorem is a standard input; a modern exposition is [10]. We cite it rather than reproduce its proof. We make no priority claim for the standard tools or the Hermite construction. Section 7 substitutes the bounds into known algorithmic tradeoffs.

## 2 Polynomial products and finite constructions

A tensor is a trilinear form in three groups of variables, called its *legs*. A *restriction* makes linear substitutions independently on the three legs. A *degeneration* is a limit of such restrictions: the substitutions may depend on  $\varepsilon \neq 0$ , and every resulting coefficient converges as  $\varepsilon \rightarrow 0$ . Our constructions first change bases and then multiply variables by integer powers of  $\varepsilon$ . The resulting coefficient array is a polynomial in  $\varepsilon$  whose constant coefficient is the desired tensor. We write  $T \geq S$  for a restriction and  $T \supseteq S$  for a degeneration. Successive constructions can use a single parameter: replace the earlier parameter by a sufficiently large power of the later one, so every earlier error term still has positive total exponent.

The *rank*  $\text{rank}(T)$  is the least number of products of three linear forms, one per leg, summing to  $T$ . The direct sum  $S \oplus T$  uses disjoint variables on all three legs and represents independent computations. The tensor product  $S \otimes T$  pairs their indices, as in a Kronecker product, and represents their composition on blocks. We write  $mT$  for  $m$  independent copies, and  $T^{\otimes N}$  for  $N$  tensor factors. Thus  $\text{rank}(T)xyz \geq T$ . All factors used below restrict to  $xyz$  by keeping one nonzero term; this is what it means to drop a factor.

The oriented dot products are

$$B_X(q) = x \sum_{j=1}^q y_j z_j,$$

with  $B_Y(q), B_Z(q)$  obtained by permuting the legs. They have rank  $q$ , and  $B_i(q) \otimes B_i(q')$  is  $B_i(qq')$  after reindexing. The matrix multiplication tensor factors as

$$\langle A, B, C \rangle = \sum_{u,v,w} x_{uv} y_{vw} z_{wu} \cong B_Y(A) \otimes B_Z(B) \otimes B_X(C). \quad (2)$$

Thus the three dot-product dimensions specify the matrix shape.

For distinct input legs  $i, j$ , with remaining output leg  $k$ , the *convolution tensor* is

$$C_{ij}(r, s) = \sum_{h=0}^{r-1} \sum_{\ell=0}^{s-1} x_h^{(i)} x_\ell^{(j)} x_{h+\ell}^{(k)}.$$

It multiplies two polynomials represented by their coefficient vectors. Its rank is  $r + s - 1$ : evaluation at  $r + s - 1$  points and interpolation give the upper bound, and the independent output coefficients give the matching matrix-rank lower bound. Decreasing either input length is a restriction. Also

$$C_{ij}(1, s) \cong B_i(s), \quad C_{ij}(r, 1) \cong B_j(r). \quad (3)$$

Blocks  $T_h$  are *matched sectors sharing leg  $i$*  when  $T = \sum_h T_h$  and the blocks may share variables on leg  $i$ , but use pairwise disjoint variables on each other leg. The following constructions hold for every permutation of the legs.

**Lemma 2.1** ([6]). *1. If  $T$  has  $M$  matched sectors sharing leg  $i$ , then*

$$(3M)T \supseteq \bigoplus_{h=1}^M (T_h \otimes B_i(M)). \quad (4)$$

*The sectors on the right are independent on all three legs.*

*2. For  $r \geq 1$  and  $s \geq 2$ ,  $C_{ij}(r, s) \otimes B_i(2)$  degenerates to two matched sectors sharing leg  $i$ , namely  $C_{ij}(r, s+1)$  and  $C_{ij}(r, s-1)$ .*

3. For  $r, s \geq 1$ ,  $C_{ij}(r, 3s + r - 1)$  degenerates to three matched sectors sharing leg  $i$ : two copies of  $C_{ij}(r, s)$  and one copy of  $C_{ik}(r, s)$ .

Appendix A gives the two polynomial splittings in coefficient coordinates. Appendix C proves (4) using a finite geometric sum and a nonnegative square. The next construction splits a convolution into fully independent smaller convolutions.

**Lemma 2.2** (Hermite direct sums). *For any nonempty finite list of positive integer pairs  $(r_h, s_h)$ , with  $R = \sum_h r_h$  and  $S = \sum_h s_h$ ,*

$$C_{ij}(R, S) \supseteq \bigoplus_h C_{ij}(r_h, s_h). \quad (5)$$

The summands are disjoint on all three legs.

*Proof.* It suffices to use the input legs  $X, Y$ . Choose distinct integers  $\alpha_1, \dots, \alpha_m$ . Hermite interpolation [9, Section 5.6] gives a basis  $u_{h,i}(t)$  for polynomials of degree less than  $R$ , specified by

$$u_{h,i}(t) \equiv \begin{cases} (t - \alpha_h)^i, & \ell = h, \\ 0, & \ell \neq h, \end{cases} \pmod{(t - \alpha_\ell)^{r_\ell}}, \quad 0 \leq i < r_h.$$

Choose the analogous basis  $v_{h,j}(t)$  for the second input, using the orders  $s_h$ . For the output polynomial  $W(t)$ , retain the coefficients  $[z^k]W(\alpha_h + z)$  for  $0 \leq k \leq r_h + s_h - 2$ ; here  $[z^k]$  means the coefficient of  $z^k$ . These  $R + S - m$  values can be specified independently: Hermite interpolation realizes any such list with a polynomial of degree less than  $R + S - m$ , which fits the output degree bound  $R + S - 1$ .

In these fixed coordinates, scale each tensor variable by  $\varepsilon$  raised to its weight, where

$$\begin{aligned} w(x_{h,i}) &= r_h - 1 - i, & w(y_{h,j}) &= s_h - 1 - j, \\ w(z_{h,k}) &= k - r_h - s_h + 2. \end{aligned} \quad (6)$$

For two inputs from node  $h$ , the coefficient at  $k = i + j$  equals one and has total weight zero. Any other nonzero coefficient at that node has  $k > i + j$  and positive weight. If the first input comes from another node and the second is  $v_{h,j}$ , a nonzero coefficient requires  $k \geq r_h + j$ . Its weight is at least one, since the first-input weight is nonnegative. The reversed case is identical. If both inputs come from other nodes, their product vanishes to order at least  $r_h + s_h$ , beyond the retained output range.

Thus every nonzero term has nonnegative weight, and the limit as  $\varepsilon \rightarrow 0$  is exactly the direct sum in (5).  $\square$

### 3 Tensor characters and counting

#### 3.1 Strassen's spectral theorem

A *tensor character* is a function  $\lambda$  from tensors to nonnegative real numbers satisfying

$$\begin{aligned} \lambda(S \oplus T) &= \lambda(S) + \lambda(T), & \lambda(S \otimes T) &= \lambda(S)\lambda(T), \\ \lambda(xyz) &= 1, & S \geq T &\implies \lambda(S) \geq \lambda(T). \end{aligned} \quad (7)$$

These rules express addition of independent computations, multiplication of tensor factors, and monotonicity under linear substitutions. Matrix rank after grouping one leg against the other two is an example. Keeping one nonzero term and using  $\text{rank}(T)xyz \geq T$  gives  $1 \leq \lambda(T) \leq \text{rank}(T)$  for every nonzero  $T$ .

Characters also cannot increase under our polynomial degenerations. If  $T \supseteq S$  has degree  $L$ , interpolation of its  $N$ -th tensor power at  $NL + 1$  nonzero points gives

$$(NL + 1)T^{\otimes N} \geq S^{\otimes N}, \quad \lambda(S) \leq (NL + 1)^{1/N} \lambda(T).$$

Indeed, each specialized tensor is a restriction of  $T^{\otimes N}$ , and interpolation extracts the constant coefficient from their direct sum. Letting  $N \rightarrow \infty$  proves  $\lambda(S) \leq \lambda(T)$ . No continuity assumption on  $\lambda$  is needed.

The *asymptotic rank* of  $T$  is

$$\tilde{R}(T) = \lim_{N \rightarrow \infty} \text{rank}(T^{\otimes N})^{1/N} = \inf_{N \geq 1} \text{rank}(T^{\otimes N})^{1/N}.$$

The limit exists because tensor rank is submultiplicative; apply the subadditive-sequence form of Fekete's lemma to its logarithm.

**Theorem 3.1** (Strassen's spectral theorem). *For every nonzero tensor  $T$  over  $\mathbb{C}$ ,*

$$\tilde{R}(T) = \max_{\lambda} \lambda(T), \tag{8}$$

where the maximum ranges over all tensor characters.

We use [7, Theorem 3.8]; see also [10, Corollary 3.30]. The easy direction follows from  $\lambda(T)^N = \lambda(T^{\otimes N}) \leq \text{rank}(T^{\otimes N})$ . The theorem supplies the opposite direction. Thus, to prove an upper bound on asymptotic rank, it suffices to bound every character value.

Fix an arbitrary character  $\lambda$ . Its *dot-product parameters* are  $p_i = \log_2 \lambda(B_i(2)) \in [0, 1]$ , for  $i = X, Y, Z$ . They determine the values at all integer dimensions:

$$\lambda(B_i(q)) = q^{p_i} \quad (q \geq 1). \tag{9}$$

To see this, put  $m = \lfloor N \log_2 q \rfloor$ . Monotonicity in dimension and  $B_i(q)^{\otimes N} \cong B_i(q^N)$  give

$$2^{mp_i} \leq \lambda(B_i(q))^N \leq 2^{(m+1)p_i}.$$

Take  $N$ -th roots and let  $N \rightarrow \infty$ . The interval  $[0, 1]$  follows from  $1 \leq \lambda(B_i(2)) \leq 2$ . In particular, (2) gives

$$\lambda(\langle A, B, C \rangle) = A^{p_Y} B^{p_Z} C^{p_X}. \tag{10}$$

Our aim is to constrain these three parameters for every character. Define its *convolution profiles* by

$$f_{ij}(r, s) = \lambda(C_{ij}(r, s)). \tag{11}$$

### 3.2 Words with prescribed multiplicities

Suppose  $T$  splits into  $s$  matched sectors  $T_1, \dots, T_s$  sharing leg  $i$ . In  $T^{\otimes N}$  a sector is specified by a word of  $N$  sector labels. Retain on the two nonshared legs the words with  $n_h$  occurrences of label  $h$ , where  $\sum_h n_h = N$ . There are

$$M = \binom{N}{n_1, \dots, n_s}$$

matched sectors. For example, multiplicities  $(2, 1)$  give the words 112, 121, 211. Each sector is a product of the  $T_h$  with the prescribed multiplicities. Apply (4); now all legs distinguish the words, so their factors can be reordered independently. This gives

$$(3M)T^{\otimes N} \supseteq M \left( \bigotimes_h T_h^{\otimes n_h} \right) \otimes B_i(M). \tag{12}$$

**Lemma 3.2** (Multinomial power-sum inequality). *If  $T \supseteq \sum_{h=1}^s T_h$  is a splitting into matched sectors sharing leg  $i$  and  $p_i > 0$ , then*

$$\lambda(T)^{1/p_i} \geq \sum_h \lambda(T_h)^{1/p_i}. \quad (13)$$

*For fully independent summands,  $T \supseteq \bigoplus_h T_h$  gives  $\lambda(T) \geq \sum_h \lambda(T_h)$  directly by additivity.*

*Proof.* For every tuple  $\sum_h n_h = N$ , applying  $\lambda$  to (12) gives

$$3M\lambda(T)^N \geq M^{1+p_i} \prod_h \lambda(T_h)^{n_h}.$$

Put  $a_h = \lambda(T_h)^{1/p_i}$ . Cancel  $M$  and take the power  $1/p_i$ :

$$M \prod_h a_h^{n_h} \leq 3^{1/p_i} \lambda(T)^{N/p_i}.$$

Sum over all  $\binom{N+s-1}{s-1}$  multiplicity tuples. The multinomial theorem gives

$$\left( \sum_h a_h \right)^N \leq \binom{N+s-1}{s-1} 3^{1/p_i} \lambda(T)^{N/p_i}.$$

For fixed  $s$ , the binomial coefficient is polynomial in  $N$ . Taking  $N$ -th roots and letting  $N \rightarrow \infty$  proves (13).  $\square$

Rank bounds and restrictions give

$$1 \leq f_{ij}(r, s) \leq r + s - 1, \quad f_{ij}(r, s) = f_{ji}(s, r), \quad (14)$$

and coordinatewise monotonicity. The equality merely interchanges the two polynomial inputs of the same convolution.

**Lemma 3.3** (Powered concavity and tripling). *For  $p_i > 0$ , the sequence  $s \mapsto f_{ij}(r, s)^{1/p_i}$  is concave, and for all integers  $r, s \geq 1$ ,*

$$f_{ij}(r, 3s + r - 1)^{1/p_i} \geq 2f_{ij}(r, s)^{1/p_i} + f_{ik}(r, s)^{1/p_i}, \quad (15)$$

*where  $i, j, k$  are distinct. For  $p_j > 0$ , the sequence  $r \mapsto f_{ij}(r, s)^{1/p_j}$  is also concave.*

*Proof.* The determinant splitting and Lemma 3.2 give, for  $s \geq 2$ ,

$$2f_{ij}(r, s)^{1/p_i} = \lambda(C_{ij}(r, s) \otimes B_i(2))^{1/p_i} \geq f_{ij}(r, s+1)^{1/p_i} + f_{ij}(r, s-1)^{1/p_i}.$$

This is discrete concavity in  $s$ ; input interchange gives the other coordinate. The three-sector splitting and the same lemma give (15).  $\square$

## 4 Ordinary scaling limits

The Hermite splitting and the full-sum assertion of Lemma 3.2 give

$$f_{ij}(r + u, s + v) \geq f_{ij}(r, s) + f_{ij}(u, v). \quad (16)$$

Taking  $m$  copies of  $C_{ij}(1, 1)$  also gives  $f_{ij}(m, m) \geq m$ ; by monotonicity,  $f_{ij}(r, s) \geq \min\{r, s\}$ .

**Lemma 4.1** (Ordinary homogeneous limits). *For every  $r, s > 0$ , the limit*

$$g_{ij}(r, s) = \lim_{n \rightarrow \infty} \frac{f_{ij}(\lfloor nr \rfloor, \lfloor ns \rfloor)}{n} \quad (17)$$

*exists, uniformly for  $r, s \in [u, v]$  for every fixed  $0 < u < v$ . The profiles  $g_{ij}$  are nondecreasing and concave, and are homogeneous of degree one, meaning*

$$g_{ij}(tr, ts) = t g_{ij}(r, s) \quad (t > 0).$$

*They also satisfy*

$$\min\{r, s\} \leq g_{ij}(r, s) \leq r + s, \quad g_{ij}(r, s) = g_{ji}(s, r). \quad (18)$$

*For  $p_i > 0$ ,  $g_{ij}^{1/p_i}$  is concave in  $s$  and satisfies*

$$g_{ij}(r, 3s + r)^{1/p_i} \geq 2g_{ij}(r, s)^{1/p_i} + g_{ik}(r, s)^{1/p_i}. \quad (19)$$

*For  $p_j > 0$ ,  $g_{ij}^{1/p_j}$  is concave in  $r$ .*

*Proof.* Fix positive integers  $r, s$ . By (16), the sequence  $a_n = f_{ij}(nr, ns)$  is superadditive and bounded above by  $n(r + s)$ . For each fixed  $k$ , writing  $n = qk + t$ ,  $0 \leq t < k$ , gives  $a_n \geq qa_k$ . Therefore

$$\liminf_{n \rightarrow \infty} \frac{a_n}{n} \geq \frac{a_k}{k}, \quad \lim_{n \rightarrow \infty} \frac{a_n}{n} = \sup_{k \geq 1} \frac{a_k}{k}.$$

This is the superadditive-sequence argument, also called Fekete's lemma.

For positive rational  $r = a/q$ ,  $s = b/q$ , monotonicity squeezes the floored pair in (17) between  $\lfloor n/q \rfloor(a, b)$  and  $(\lfloor n/q \rfloor + 1)(a, b)$ . Hence the limit is  $g_{ij}(a, b)/q$ . Superadditivity passes to rational pairs, and rational homogeneity makes the limit concave there.

To handle real lengths, first bound how much the limit can change between nearby rational points. Fix rational  $0 < u < v$  and write  $g = g_{ij}$ . For rational  $r, r', s \in [u, v]$  with  $r < r'$ , concavity gives

$$0 \leq \frac{g(r', s) - g(r, s)}{r' - r} \leq \frac{g(r, s) - g(u/2, s)}{r - u/2} \leq \frac{4v}{u},$$

using  $0 \leq g(r, s) \leq r + s \leq 2v$ . The same bound holds in the other coordinate. Thus changing the two coordinates by a total of  $\eta$  changes  $g$  by at most  $4v\eta/u$ . Approximating real lengths by rationals therefore defines a unique continuous extension of  $g$ .

Now cover any fixed rectangle with a fine rational grid. The slope bound makes the limit values at neighboring corners arbitrarily close. Both the normalized discrete profiles and  $g$  lie between their values at the lower and upper grid corners. Convergence at the finitely many corners therefore gives uniform convergence throughout the rectangle. Homogeneity, monotonicity, input interchange, and concavity, as well as (18), pass from rational to real lengths by continuity.

For powered concavity, first take rational lengths and a rational interpolation coefficient. Clear denominators and apply the discrete concavity from Lemma 3.3 along multiples of that common denominator. Divide by  $n^{1/p_i}$  and pass to the limit; continuity gives all real lengths and coefficients. The other coordinate is identical. Finally, local uniform convergence absorbs the bounded rounding errors and the  $-1$  in (15), giving (19).  $\square$

## 5 Common slopes and the angular constraint

Index the three tensor legs by 1, 2, 3. We use the limits above to constrain the three dot-product parameters without averaging their orientations.

**Lemma 5.1** (Common slope and intercepts). *Fix  $i$  with  $p_i > 0$ , and let  $j, k$  be the other legs. The functions  $P_{ij}(s) = g_{ij}(1, s)^{1/p_i}$  and  $P_{ik}(s) = g_{ik}(1, s)^{1/p_i}$  have a common asymptotic slope  $L_i > 0$ . Their intercepts*

$$C_{ij} = \lim_{s \rightarrow \infty} (P_{ij}(s) - L_i s), \quad C_{ik} = \lim_{s \rightarrow \infty} (P_{ik}(s) - L_i s)$$

are finite and nonnegative, with  $C_{ij} + C_{ik} \leq L_i$ .

*Proof.* Write  $P = P_{ij}$  and  $Q = P_{ik}$ . Positive nondecreasing concavity gives finite nonnegative limits  $L_P = \lim P(s)/s$  and  $L_Q = \lim Q(s)/s$ . Dividing the coupled inequalities

$$P(3s+1) \geq 2P(s) + Q(s), \quad Q(3s+1) \geq 2Q(s) + P(s)$$

by  $s$  and passing to infinity gives  $L_P \geq L_Q$  and  $L_Q \geq L_P$ . Call their common value  $L_i$ .

Their sum  $R = P + Q$  satisfies  $R(3s+1) \geq 3R(s)$ . Iterating at  $s_m = 3^m(s + \frac{1}{2}) - \frac{1}{2}$  gives

$$R(s) \leq \lim_{m \rightarrow \infty} 3^{-m} R(s_m) = L_i(2s+1). \quad (20)$$

Since  $R(s) > 0$ , we have  $L_i > 0$ . For  $v > u > 0$ , concavity bounds the secant slope  $(P(v) - P(u))/(v - u)$  below by every later secant slope; sending its far endpoint to infinity gives

$$P(v) - P(u) \geq L_i(v - u).$$

Thus  $P(s) - L_i s$  is nondecreasing and nonnegative, using the nonnegative limit of  $P$  at zero. The same holds for  $Q$ . Their sum is bounded above by  $L_i$  by (20), so their limits at infinity exist and have the stated bounds.  $\square$

**Proposition 5.2** (Angular constraint). *The dot-product parameters of every tensor character satisfy*

$$\sum_{i=1}^3 \arccos \sqrt{p_i} \geq \frac{\pi}{2}. \quad (21)$$

*Proof.* If some  $p_i = 0$ , its summand is already  $\pi/2$ . Assume all three parameters are positive. Set  $c_{ij} = C_{ij}/L_i$ . Lemma 5.1 gives the six intercept ratios and their three bounds

$$c_{ij} \geq 0, \quad c_{ij} + c_{ik} \leq 1. \quad (22)$$

The secant bound from Lemma 5.1 and the definition of  $c_{ij}$  give, for  $s, t > 0$ ,

$$P_{ij}(s+t) \geq P_{ij}(s) + L_i t, \quad P_{ij}(s) \leq L_i(s + c_{ij}).$$

Divide these inequalities and use homogeneity to obtain

$$\frac{g_{ij}(r, s+t)}{g_{ij}(r, s)} \geq \left(1 + \frac{t}{s + c_{ij}r}\right)^{p_i}. \quad (23)$$

Increase  $r$  by  $\varepsilon r$ , then  $s$  by  $\varepsilon s$ . Input interchange supplies the first bound and (23) supplies the second:

$$\begin{aligned} 1 + \varepsilon &= \frac{g_{ij}((1+\varepsilon)r, (1+\varepsilon)s)}{g_{ij}(r, s)} \\ &\geq \left(1 + \frac{\varepsilon r}{r + c_{ji}s}\right)^{p_j} \left(1 + \frac{\varepsilon s}{s + c_{ij}(1+\varepsilon)r}\right)^{p_i}. \end{aligned}$$

Take logarithms, divide by  $\varepsilon$ , and let  $\varepsilon \downarrow 0$ . The elementary limit  $\log(1 + \varepsilon x)/\varepsilon \rightarrow x$  gives

$$1 \geq \frac{p_i s}{s + c_{ij} r} + \frac{p_j r}{r + c_{ji} s} \quad (r, s > 0). \quad (24)$$

With  $t = r/s$ , this is equivalent to

$$(1 - p_j)c_{ij}t^2 + (1 - p_i - p_j + c_{ij}c_{ji})t + (1 - p_i)c_{ji} \geq 0 \quad (t > 0).$$

A quadratic  $At^2 + Bt + C$  with  $A, C \geq 0$  is nonnegative on  $t > 0$  exactly when  $B \geq -2\sqrt{AC}$ , including zero coefficients. Thus

$$\sqrt{c_{ij}c_{ji}} \geq K_{ij} := \left( \sqrt{p_i p_j} - \sqrt{(1 - p_i)(1 - p_j)} \right)_+, \quad (25)$$

where  $(z)_+ = \max\{z, 0\}$ .

Let  $K$  be the symmetric matrix with these off-diagonal entries and zero diagonal. For every vector  $x \geq 0$ , AM–GM and (22) give

$$x^\top K x \leq \sum_{i < j} 2\sqrt{c_{ij}c_{ji}}x_i x_j \leq \sum_{i < j} (c_{ij}x_i^2 + c_{ji}x_j^2) \leq \sum_i x_i^2. \quad (26)$$

Put  $\theta_i = \arccos \sqrt{p_i}$ . If  $\sum_i \theta_i < \pi/2$ , increase all three angles by the same positive amount to obtain  $\phi_i > 0$  with sum  $\pi/2$ . The matrix  $K'$  with zero diagonal and  $K'_{ij} = \cos(\phi_i + \phi_j) = \sin \phi_k$  is strictly smaller than  $K$  off the diagonal. But the positive vector  $x_i = \cos \phi_i$  satisfies

$$(K'x)_i = \sin \phi_k \cos \phi_j + \sin \phi_j \cos \phi_k = \cos \phi_i = x_i,$$

contradicting (26).  $\square$

## 6 From character bounds to running times

Strassen's theorem turns the constraint on dot-product parameters into rank bounds. We first spell out how these bounds give algorithms.

**Lemma 6.1** (Character bounds give algorithms). *Fix  $\mathbf{h} = (h_X, h_Y, h_Z) \geq 0$ . If the dot-product parameters of every tensor character satisfy  $\sum_i h_i p_i \leq E$ , then  $\omega(h_Y, h_Z, h_X) \leq E$ .*

*Proof.* The zero target is trivial. For a fixed integer  $N \geq 2$ , put  $D_i = \lceil N^{h_i} \rceil$  and  $M_N = \bigotimes_i B_i(D_i)$ . Since  $D_i \leq 2N^{h_i}$  and  $0 \leq p_i \leq 1$ , every character satisfies

$$\lambda(M_N) = \prod_i D_i^{p_i} \leq 8N^{\sum_i h_i p_i} \leq 8N^E.$$

Theorem 3.1 therefore gives

$$\tilde{R}(M_N) \leq 8N^E. \quad (27)$$

Fix any positive exponent slack  $\eta$ . First choose  $N$  so large that  $8 < N^{\eta/2}$ . The infimum defining  $\tilde{R}(M_N)$  then gives a fixed integer  $K$  and an exact rank decomposition with

$$R := \text{rank}(M_N^{\otimes K}) \leq N^{K(E+\eta)}.$$

This is a finite bilinear algorithm for matrix dimensions  $(D_Y^K, D_Z^K, D_X^K)$ . Apply it recursively  $\ell$  times to blocks. By (2), the resulting dimensions are  $(D_Y^{K\ell}, D_Z^{K\ell}, D_X^{K\ell})$  and there are  $R^\ell$  scalar multiplications.

Each input and output dimension, viewed as a number of scalar entries, is at most  $R$ , by the corresponding matrix-rank lower bound. Thus the linear substitutions and additions over all recursion levels use  $O(\ell R^\ell)$  operations. Given input size  $n$ , choose  $\ell = \lceil \log_N n/K \rceil$  and pad to these dimensions. They cover the target shape because  $D_i^{K\ell} \geq n^{h_i}$ , while  $R^\ell = O(n^{E+\eta})$ . This proves the bound for every positive slack, and hence the lemma under our exponent convention. All choices of  $N$ ,  $K$ , and the finite identity precede the limit  $n \rightarrow \infty$ .  $\square$

We now optimize the angular constraint for three arbitrary dimensions; the rectangular curve follows by substitution.

**Corollary 6.2.** *For every fixed  $a \geq b \geq c \geq 0$ ,*

$$\omega(a, b, c) \leq \begin{cases} a + b, & c(a + b) \leq ab, \\ \frac{a + b + c}{2} + \frac{1}{4} \left( \frac{ab}{c} + \frac{ac}{b} + \frac{bc}{a} \right), & c(a + b) > ab. \end{cases} \quad (28)$$

*When  $c(a + b) \leq ab$ , equality holds by the size lower bound.*

*Proof.* Reorder the character's dot-product parameters so that the objective is  $ap_1 + bp_2 + cp_3$ , and write  $p_i = \cos^2 \theta_i$  with  $\theta_i \in [0, \pi/2]$ . By Proposition 5.2, their sum is at least  $\pi/2$ . Since the objective decreases with each angle, its maximum over this region occurs on  $\sum_i \theta_i = \pi/2$ . If a weight is zero, the maximum is  $a + b$ . Assume  $a, b, c > 0$ .

On any boundary edge, the other two angles sum to  $\pi/2$ , so their contribution is a convex combination of the corresponding two weights. The largest boundary value is therefore  $a + b$ . At an interior stationary point, the multiplier equations are

$$a \sin(2\theta_1) = b \sin(2\theta_2) = c \sin(2\theta_3).$$

The angles  $2\theta_i$  form a triangle with sides proportional to  $1/a, 1/b, 1/c$ . Such a triangle exists uniquely exactly when  $1/c < 1/a + 1/b$ , or  $c(a + b) > ab$ . The cosine rule gives

$$\cos(2\theta_1) = \frac{c}{2b} + \frac{b}{2c} - \frac{bc}{2a^2},$$

and cyclic variants. Using  $p_i = (1 + \cos(2\theta_i))/2$  yields the second branch of (28). It exceeds  $a + b$  by  $(c(a + b) - ab)^2 / (4abc) > 0$ . Thus it is the global maximum when the triangle exists; otherwise the maximum is the boundary value  $a + b$ . At  $c(a + b) = ab$ , the branches agree.

We have bounded the objective for every tensor character. Lemma 6.1, with target  $\mathbf{h} = (c, a, b)$ , gives (28).  $\square$

*Proof of Theorem 1.1.* Apply Corollary 6.2 to the sorted dimensions of  $(1, k, 1)$ . For  $0 \leq k \leq 1$ , the first branch applies exactly when  $2k \leq 1$ . In the remaining range, both possible orderings give

$$\frac{2 + k}{2} + \frac{1}{4} \left( 2k + \frac{1}{k} \right) = 1 + k + \frac{1}{4k}.$$

The output-size lower bound  $\omega(k) \geq 2$  proves equality for  $0 \leq k \leq \frac{1}{2}$  and hence  $\alpha \geq \frac{1}{2}$ .  $\square$

Some values of the bound in Theorem 1.1 are:

| $k$              | 0.5 | 0.6       | 0.8    | 1    | 1.5       | 2     |
|------------------|-----|-----------|--------|------|-----------|-------|
| $\omega(k) \leq$ | 2   | 2.0166... | 2.1125 | 2.25 | 2.6666... | 3.125 |

The exponent convention gives an  $O(n^2)$ -time algorithm at  $k = \frac{1}{2}$ .

## 7 Applications

Theorem 1.1 and Corollary 6.2 give the following bounds through standard rectangular-multiplication balances. All bounds use arithmetic operations, with graph operations and symbol comparisons also charged.

1. **One lowest common ancestor per pair in a DAG.** Grandoni et al. [3, Theorem 4.2 of the published version] obtain exponent  $1+2x$  from the balance  $\omega(x) = 3x$ . Theorem 1.1 permits  $x = (1+\sqrt{3})/4$ , giving  $O(n^{(3+\sqrt{3})/2})$  time for finding one lowest common ancestor for every pair in an  $n$ -vertex DAG. The exponent is  $(3 + \sqrt{3})/2 \approx 2.36603$ .
2. **Exact all-pairs Hamming distance.** For  $n$  vectors of length  $n$  over an unrestricted alphabet, the reduction to dominance products [5] and Yuster’s tradeoff [11, Proposition 2.1] give the balance  $\rho = \omega(4 - \rho)$ . Theorem 1.1 gives  $\rho = (13 - \sqrt{7})/4 \approx 2.58856$  and running time  $O(n^\rho)$  for computing every exact Hamming distance.
3. **The standard unweighted APSP balance.** Zwick’s deterministic algorithm for directed unweighted APSP [13, Section 6] has exponent  $2 + \mu$ , where  $\omega(\mu) \leq 1 + 2\mu$ . Here  $\omega(\frac{1}{2}) = 2 = 1 + 2(\frac{1}{2})$ , so it gives  $O(n^{2.5})$  running time. Combining the full curve with the deterministic (min, +)-product algorithm and APSP reduction of Alman and Vassilevska Williams [1, Theorems 22 and 33–34] further gives  $O(n^{2.4999})$  running time.
4. **Minimum and maximum witnesses.** For two Boolean  $n \times n$  matrices, a minimum (maximum) witness for entry  $(i, j)$  is the least (greatest) index  $k$  with  $A_{ik} = B_{kj} = 1$ . The standard blocking algorithm [2, Section 2.1] takes  $O(n^{1-r+\omega(r)} + n^{2+r})$  time. Taking  $r = \frac{1}{2}$  gives  $O(n^{2.5})$  time for all witnesses.
5. **Sparse matrix multiplication.** For two  $n \times n$  matrices with at most  $m$  nonzero entries each, the sparse–dense decomposition of Yuster and Zwick [12, Section 3] gives  $O(m^2/n^r + n^{\omega(r)})$  time for  $0 \leq r \leq 1$ . Thus  $m \leq n^{1.25}$  permits  $O(n^2)$  time.
6. **Dynamic matrix inverse and reachability.** Substituting Corollary 6.2 into van den Brand–Nanongkai–Saranurak’s tradeoff [8, Theorem 4.2], with  $u = 0.50323551$  and  $v = 0.80667763$  in  $\max\{u + v, \omega(1, u, v) - u, \omega(v) - v\}$ , gives  $O(n^{1.30992})$  worst-case time per entry update and query for maintaining the inverse of a nonsingular  $n \times n$  matrix, after  $O(n^{2.25})$  preprocessing. The associated Monte Carlo reduction gives the same bounds for directed reachability under edge updates, with  $O(1)$  query time for a fixed pair [8, Corollary C.12 and Theorem C.16].

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## A The determinant construction

We recap the determinant construction used in [6, Lemma 4.1] as an explicit change of basis for pairs of univariate polynomials. Its classical form is the degree-one Clebsch–Gordan formula [4, Theorem 2.6.3, pp. 56–57 of the author version].

Fix  $r \geq 1$  and  $s \geq 2$ . The tensor  $C_{XY}(r, s) \otimes B_X(2)$  computes

$$(f, (a, b)) \mapsto (fa, fb), \quad \deg f < r, \quad \deg a, \deg b < s.$$

Thus tensoring with  $B_X(2)$  means multiplying two polynomials by the same  $f$ . We will choose fixed bases in which this computation splits, in the limit, into multiplication with input lengths  $s + 1$  and  $s - 1$ .

For  $e \geq 1$ , consider pairs  $(a, b)$  with  $\deg a, \deg b \leq e$ . The polynomial  $a + tb$  has degree at most  $e + 1$ , and

$$a + tb = 0 \iff (a, b) = (-th, h) \text{ for some } \deg h \leq e - 1. \quad (\text{A.1})$$

Use the following basis, arranged in two blocks:

$$\underbrace{(t^j, 0) \ (0 \leq j \leq e), \ (0, t^e)}_{e+2 \text{ vectors}}, \quad \underbrace{(-t^{j+1}, t^j) \ (0 \leq j < e)}_{e \text{ vectors}}.$$

The first block maps under  $(a, b) \mapsto a + tb$  to  $1, t, \dots, t^{e+1}$ . Every pair mapping to zero is a unique linear combination of the second block, by (A.1). These vectors therefore form a basis. At  $e = s - 1$ , the two blocks have sizes  $s + 1$  and  $s - 1$ .

Choose these bases for both the second-input pairs ( $e = s - 1$ ) and the output pairs ( $e = r + s - 2$ ). Multiplication by every  $f$  satisfies

$$fa + t(fb) = f(a + tb), \quad f(-th, h) = (-t(fh), fh).$$

The first identity says that the first block undergoes ordinary multiplication by  $f$ ; the second says the same for the second block and shows that it contributes nothing to the first. Consequently, the matrix of multiplication by  $f$  has the form

$$\begin{pmatrix} M_f^+ & 0 \\ R_f & M_f^- \end{pmatrix}, \quad \begin{matrix} M_f^+ : \mathbb{C}^{s+1} \longrightarrow \mathbb{C}^{r+s}, \\ M_f^- : \mathbb{C}^{s-1} \longrightarrow \mathbb{C}^{r+s-2}, \end{matrix} \quad (\text{A.2})$$

where  $M_f^+, M_f^-$  are the usual coefficient matrices for multiplying by  $f$ . The bases are fixed independently of  $f$ , so this block structure holds for the whole tensor.

Scale the variables in the second block on the second-input leg by  $\varepsilon^{-1}$ , and those in the second block on the output leg by  $\varepsilon$ . On every multiplication matrix this gives

$$\begin{pmatrix} I & 0 \\ 0 & \varepsilon I \end{pmatrix} \begin{pmatrix} M_f^+ & 0 \\ R_f & M_f^- \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & \varepsilon^{-1} I \end{pmatrix} = \begin{pmatrix} M_f^+ & 0 \\ \varepsilon R_f & M_f^- \end{pmatrix} \xrightarrow{\varepsilon \rightarrow 0} \begin{pmatrix} M_f^+ & 0 \\ 0 & M_f^- \end{pmatrix}.$$

The two surviving tensor blocks are  $C_{XY}(r, s + 1)$  and  $C_{XY}(r, s - 1)$ . They share the first leg and have disjoint second and third legs. Permuting the legs gives the other orientations.

## B The three-sector splitting

We recap the construction from [6, Lemma 4.2]. In  $C_{XY}(r, 3s + r - 1)$ , a term has  $0 \leq x < r$  and output index  $z = x + y$ . Partition the second-input and output indices into three intervals:

|   | left             | middle                    | right                       |
|---|------------------|---------------------------|-----------------------------|
| Y | $[0, s - 1]$     | $[s, 2s + r - 2]$         | $[2s + r - 1, 3s + r - 2]$  |
| Z | $[0, s + r - 2]$ | $[s + r - 1, 2s + r - 2]$ | $[2s + r - 1, 3s + 2r - 3]$ |

Scale middle  $Y$  variables by  $\varepsilon$  and middle  $Z$  variables by  $\varepsilon^{-1}$ , leaving the others unchanged. Since  $z = y + x$  with  $0 \leq x < r$ , left and right  $Y$  indices stay in the corresponding  $Z$  intervals. Cross-sector terms thus connect middle  $Y$  to outer  $Z$  and acquire a factor  $\varepsilon$ ; matched terms remain unchanged. As  $\varepsilon \rightarrow 0$ , precisely three matched sectors sharing  $X$  remain.

Translations identify the outer sectors with  $C_{XY}(r, s)$ . The middle is

$$\sum_{u=0}^{r-1} \sum_{v=0}^{s-1} x_{r-1-u} y_{s+u+v} z_{s+r-1+v},$$

which is  $C_{XZ}(r, s)$  after reversing its first-leg basis. This identifies the individual sector. In the fixed-word construction of (12), the sectors acquire independent first-leg variables before their individual bases are changed.

## C Finite separation of a shared leg

We use the construction of [6, Proposition 3.1], reducing the copy count from  $5M$  to  $3M$ . It suffices to share leg  $X$ . Write

$$T = \sum_{h=1}^M \sum_{a,b,c} \mu_{habc} x_a y_{hb} z_{hc}.$$

Take  $L = 3M$  copies and a primitive  $L$ -th root of unity  $\zeta$ . In copy  $\ell = 0, \dots, L-1$ , make the substitutions

$$\begin{aligned} x_a^{(\ell)} &\mapsto \sum_{g=1}^M \zeta^{2\ell g} X_{ag}, \\ y_{hb}^{(\ell)} &\mapsto \sum_{u=1}^M \zeta^{\ell(u-h)} Y_{hbu}, \\ z_{hc}^{(\ell)} &\mapsto \frac{1}{L} \sum_{v=1}^M \zeta^{\ell(-v-h)} Z_{hcv}. \end{aligned}$$

Summing the copies multiplies each coefficient by

$$\frac{1}{L} \sum_{\ell=0}^{L-1} \zeta^{\ell[u-v+2(g-h)]}.$$

The geometric sum is one for exponent divisible by  $L$  and zero otherwise. Since  $|u - v + 2(g - h)| \leq 3(M - 1) < L$ , it retains precisely  $u - v + 2(g - h) = 0$ .

Now give  $X_{ag}, Y_{hbu}, Z_{hcv}$  respective weights  $g^2$ ,  $hu - h^2$ , and  $-hv$ . On every retained term their total is

$$g^2 + h(u - v) - h^2 = (g - h)^2 \geq 0.$$

The weight-zero terms have  $g = h$  and  $u = v$ , so the limit is

$$\sum_{h,a,b,c,u} \mu_{habc} X_{ah} Y_{hbu} Z_{hcu} = \bigoplus_h (T_h \otimes B_X(M)).$$

Every leg now distinguishes  $h$ . Although individual weights may be negative, the transformed coefficients form a polynomial of degree at most  $(M - 1)^2$ . This proves (4).